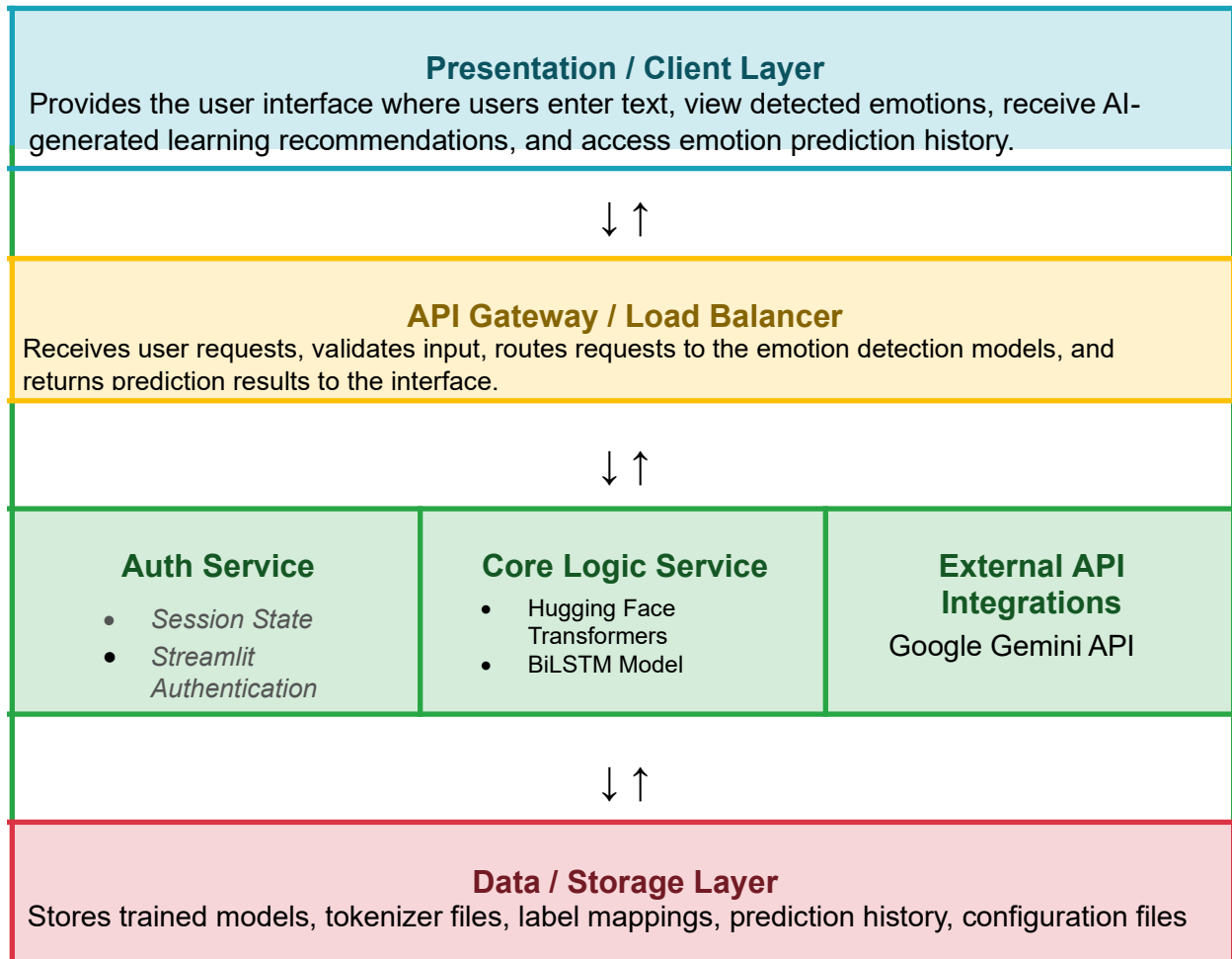


Solution Architecture

Date	10 JULY 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User interface for text input, emotion visualization, recommendation display, and history tracking	Streamlit, HTML, CSS, Python
API Gateway	Receives user input, validates requests, invokes emotion detection services, and sends results back to the interface.	<i>Streamlit Server, Python</i>
Microservices / Core Logic	Performs preprocessing, BiLSTM and BERT emotion prediction, model fusion, recommendation generation, and logging	<i>TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, Pandas, NumPy</i>
Database / Storage	Stores trained models, tokenizer, prediction history, configuration files, metrics, and label mappings.	CSV, JSON, Keras (.keras), Safetensors (.safetensors)